

Semantic extraction of PHQ-9 / GAD-7 / Epworth items to SNOMED CT

Project recap — presentation support material

1. Objective and method

Pipeline followed: questionnaire item → Qwen2.5-7B-Instruct (Ollama, local) → manual search in SNOMED CT → validation → RDF/OWL/Turtle fragment. Qwen has no knowledge of, and no access to, SNOMED CT: it proposes a concept in precise medical English, never a code. The verification and the choice of match remain entirely manual.

2. Selection of the 9 items

Chosen to cover distinct semantic categories, so Qwen would not only be tested on affective symptoms.

Instrument	Items	Rationale
PHQ-9	phq2, phq4, phq7, phq9	mood, somatic symptom, cognition, safety item (suicidal ideation)
GAD-7	gad1, gad4, gad6	general anxiety, psychophysiological manifestation, irritability
Epworth	epw4, epw8	two situations with different alertness demand (passive passenger vs. stopped in traffic)

One Epworth item (epw2) and one original item were dropped along the way to speed up the runs (from 10 down to 9 items).

3. Prompt evolution (the key point for evaluation)

Version 1 (before this work): too generic a prompt, no category constraint, no rule against inferring a diagnosis, no strictly enforced output format.

Versions 2 and 3, successive improvements:

- Closed categories (finding, symptom, disorder, behavior, observable_entity, assessment_scale) instead of free-text labels.
- Explicit prohibition on inventing a SNOMED CT code.
- Optional, typed relation (predicate from a closed list, target_concept different from the concept itself).
- "Symptom, not disorder" rule: the item must be linked to the symptom it measures, not to the disorder the questionnaire screens for.
- Epworth-specific rule: the situation (car, TV) is only a probe for sleepiness, not the measured concept itself.
- Evidence field: first an exact quote from the item (redundant with the Source column), then a one-sentence justification, without copying the item or naming an absent diagnosis.
- Status (existing_concept / candidate_project_concept) requested directly from Qwen's output, to be confirmed after manual search.

Concrete effect: better format discipline (valid JSON, categories respected), and a clear improvement on the Epworth items (risk_of_sleeping_while_driving → daytime sleepiness). However, several substantive errors persisted across versions: Qwen keeps drifting toward the "disorder" level rather than the "symptom" level for phq2, phq7 and gad4, and still produces a relation that redescribes the concept instead of linking it to the item.

4. Results and corrections per item

Item	Concept proposed by Qwen	Validated concept	Correction needed
phq2	depression	Depressed mood	Yes — Qwen targeted the disorder, not the mood state
phq4	fatigue	Fatigue	No

Item	Concept proposed by Qwen	Validated concept	Correction needed
phq7	attention_deficit	Poor concentration	Yes — drifted toward ADHD, absent from the item
phq9	suicidal ideation	Suicidal thoughts	Yes, relation only (concept correct)
gad1	anxiety	Anxiety	Yes, relation only (concept correct)
gad4	anxiety	none (Candidate Project Concept)	Yes — proposal redundant with gad1
gad6	irritability	Feeling irritable	Yes, relation only (concept correct)
epw4	risk_of_sleepiness → daytime sleepiness	Excessive daytime sleepiness	No on the final concept, label unstable across runs
epw8	risk_of_sleeping_while_driving → daytime sleepiness	Excessive daytime sleepiness	Yes — first version too specific (driving)

5. Rationale for the Match decisions

Rule applied: **Exact only if the SNOMED concept covers everything the item states.** Many PHQ-9/GAD-7 items are disjunctions ("A, B, or C"), and the SNOMED concept often covers only one of the terms.

- **phq4 (Exact):** "tired or little energy" — Fatigue covers both facets (synonym: Weariness).
- **phq7 (Exact):** "trouble concentrating" — Poor concentration matches exactly, with no added diagnosis.
- **gad6 (Exact):** "easily annoyed or irritable" — Feeling irritable covers the core meaning, minor gap tolerated.
- **phq2, gad1, phq9 (Narrower):** composite items where one term is not covered ("hopeless", "nervous"/"on edge", self-harm not always suicidal).
- **epw4, epw8 (Narrower):** Excessive daytime sleepiness adds a clinical threshold ("excessive") that the item does not presuppose, since it measures a graded probability from 0 to 3.
- **gad4 (No Match):** no SNOMED concept found for "trouble relaxing", despite several search terms tried → Candidate Project Concept.

Candidate concepts explored and rejected (worth mentioning in the presentation, as it shows the search work): Attention deficit hyperactivity disorder suspected (a suspected-diagnosis situation), Feeling nervous (too narrow for gad1), Lack of energy (too narrow for phq4), Drowsy (a state with no daytime dimension) and Hypersomnia (a disorder, not a symptom) for the Epworth items.

6. RDF/Turtle structure

Follows the required model: Questionnaire —hasItem→ QuestionnaireItem —assesses→ ClinicalConcept, with hasResponseScale kept separate. Each item, concept and scale is a distinct entity (item and concept are never merged).

Rule applied to owl:sameAs: used only for the 3 Exact matches (Fatigue, Poor concentration, Feeling irritable). For Narrower matches and the No Match case, the clinical concept still exists in the graph (with an explanatory comment) but with no equivalence link to SNOMED: the approximate alignment is recorded only in the mapping table, never asserted as equivalence in the RDF.

7. Validation

- **Syntax:** the file is loaded with rdflib (Graph().parse(..., format="turtle")) → no errors, 138 triples.
- **Semantics,** checked automatically then reviewed manually: every item has exactly one questionnaire, one concept, one scale; no item is also typed as a clinical concept; owl:sameAs appears only on the 3 intended Exact concepts; no SNOMED URI was invented (closed list of verified identifiers).

8. Limitations to acknowledge in the presentation

- The 7B model tends to drift toward the diagnosis rather than the symptom, despite an explicit rule against it.

- Composite items ("A, B, or C") structurally cause an equivalence problem: no version of the prompt solved this — it is a limitation of the one-item/one-concept mapping itself.
- Excessive daytime sleepiness remains a compromise: its more neutral parent (Daytime somnolence, mentioned in its SNOMED record) could not be found in the search.
- gad4 illustrates a genuine terminology gap in SNOMED CT for a widely used clinical item.